

Inactivation of Tautomerase Activity of Macrophage Migration Inhibitory Factor by Sulforaphane: A Potential Biomarker for Anti-inflammatory Intervention

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Background Macrophage migration inhibitory factor (MIF), a proinflammatory cytokine with keto-enol tautomerase activity, rises rapidly in response to inflammation, and is elevated in many chronic diseases. Isothiocyanates, such as sulforaphane from broccoli, are very potent inactivators of MIF tautomerase activity. A simple rapid method for determining this activity in tissues and body fluids may therefore be valuable for assessing severity of inflammation and efficacy of intervention.

Methods Existing spectrophotometric assays of MIF, based on conversion of methyl L-dopachrome to methyl 5,6-dihydroxyindole-2-carboxylate and associated loss of absorption at 475 nm, lack sensitivity. Assay sensitivity and efficiency were markedly improved by reducing the nonenzymatic rate, by lowering pH to 6.2, replacing phosphate (which catalyzes the reaction) with Bis-Tris buffer, and converting to a microtiter plate format.

Results A structure-potency study of MIF tautomerase inactivation by isothiocyanates showed that sulforaphane, benzyl, n-hexyl, and phenethyl isothiocyanates were especially potent. MIF tautomerase could be readily quantified in human urine concentrated by ultrafiltration. This activity comprised: (i) a heat-labile, sulforaphane-inactivated macromolecular fraction (presumably MIF) that was concentrated during ultrafiltration; (ii) a flow-through fraction, with constant activity during filtration, that was heat-stable, and insensitive to sulforaphane. Administration of the sulforaphane precursor glucoraphanin to human volunteers almost completely abolished urinary tautomerase activity, which was recovered over many hours.

Conclusions A simple, rapid, quantitative MIF tautomerase assay has been developed as a potential biomarker for assessing inflammatory severity and effectiveness of intervention. **Impact** An improved assay for measuring MIF tautomerase activity and its applications are described.